Jai Jalaram Packaging Web Application

1. Introduction :

1.1 purpose

- This document defines the functional and non-functional requirements of the Jai Jalaram Packaging web application.
- The system is designed to provide an online platform for packaging product sales, custom box requests, order management, invoicing, and inventory tracking.

1.2 Scope

- The system provides the following features:
- User registration with OTP verification
- Secure login with role-based access (Admin & Customer)
- Product browsing and cart-based ordering
- Custom packaging order with PDF design upload
- Admin order approval and status tracking
- Invoice generation and PDF download
- Inventory and revenue management
- Email notifications for system events

1.3 Definitions

Term	**Description**
Customer	User who places orders
Admin	User who manages system operations
Product Order	Order of standard catalog items
Custom Order	Order with custom dimensions and design

2. Description :

2.1 Product Perspective :

The system is a full-stack web application:

- Frontend: React.js
- Backend: Node.js + Express
- Database: MySQL

2.2 User Classes

Customer

- Register & login
- Browse products
- Place orders
- Track order status
- Chat with admin
- Download invoices

Admin

- Manage products & inventory
- Approve/reject orders
- Generate invoices
- Track revenue
- Manage customers

2.3 Operating Environment

- Frontend: React (Create React App)
- Backend: Node.js / Express
- Database: MySQL
- Email Service: SMTP (Nodemailer)

2.4 Constraints

- JWT authentication required
- Admin identified via predefined email list
- File upload limits:
  - PDF ≤ 5MB
  - Image ≤ 10MB
- Orders blocked if stock unavailable

2.5 Assumptions

- MySQL server is configured
- SMTP credentials available
- Environment variables configured

3. Functional Requirements

FR-1 Authentication

- User registration with OTP
- Login with email/password
- Role-based access (Admin/Customer)
- Password reset via email

FR-2 User Profile

- View profile
- Update details (name, phone, address)

FR-3 Product Catalog

- View all products
- Admin can add/update/delete products

- Low-stock alert for admin
- Admin add material

FR-4 Product Ordering

- Add items to cart
- Place multi-product order
- Automatic price calculation
- Prevent order if stock unavailable

FR-5 Custom Box Ordering

- Enter dimensions & quantity
- Upload design PDF
- Phone validation
- Price calculation based on size

FR-6 Order Management

- Admin views all orders
- Update order status
- Reduce stock after approval
- Notify customer via email

FR-7 Chat System

- Customer ↔ Admin communication
- Store messages with timestamp

FR-8 Invoice System

- Generate invoice after order completion
- Auto invoice number
- Download PDF
- Share invoice with customer

FR-10 Material Management

- Add/update/delete materials
- Track stock & cost
- Calculate inventory value

4. Data Requirements

Main Entities

- **User** → name, email, phone, role
- **Product** → name, price, stock
- **Order** → type, quantity, status
- **Invoice** → number, order_id, amount
- **Payment** → amount, method
- **Material** → quantity, price

Users Table

Stores all registered users (Admin & Customer).

Field	Type	Description
id	INT (PK)	Unique user ID
name	VARCHAR	User name
email	VARCHAR (Unique)	Email address
phone	VARCHAR	Phone number
address	TEXT	User address
password	VARCHAR	Encrypted password
role	ENUM	admin / customer
created_at	TIMESTAMP	Registration time

Products Table

Stores packaging product details.

Field	Type	Description
id	INT (PK)	Product ID
name	VARCHAR	Product name
description	TEXT	Product details
price	DECIMAL	Product price

Field	Type	Description
stock	INT	Available stock
image	LONGTEXT	Product image
created_at	TIMESTAMP	Created date

Orders Table

Stores all customer orders.

Field	Type	Description
id	INT (PK)	Order ID
user_id	INT (FK)	Linked user
order_type	ENUM	product / custom
total_price	DECIMAL	Total amount
status	ENUM	pending / approved / rejected / completed
customer_name	VARCHAR	Customer name
phone	VARCHAR	Phone
dimensions	VARCHAR	Custom size
design_file	LONGTEXT	PDF file
created_at	TIMESTAMP	Order date

Order Items Table

Stores products inside an order.

Field	Type	Description
id	INT (PK)	Item ID
order_id	INT (FK)	Order reference
product_id	INT (FK)	Product reference
quantity	INT	Quantity
price	DECIMAL	Price

Invoices Table

Stores generated invoices.

Field	Type	Description
id	INT (PK)	Invoice ID
order_id	INT (FK)	Linked order
invoice_number	VARCHAR	Unique number
total_amount	DECIMAL	Amount
shared	BOOLEAN	Shared with customer
created_at	TIMESTAMP	Date

Payments Table

Stores payment records.

Field	Type	Description
id	INT (PK)	Payment ID
order_id	INT (FK)	Order reference
amount	DECIMAL	Paid amount
method	VARCHAR	Payment type
status	VARCHAR	Payment status
created_at	TIMESTAMP	Date

Materials Table

Stores raw material inventory.

Field	Type	Description
id	INT (PK)	Material ID
name	VARCHAR	Material name
unit	VARCHAR	Unit (kg, piece)
quantity	INT	Available stock

Field	Type	Description
unit_price	DECIMAL	Price per unit
created_at	TIMESTAMP	Date

5. External Interfaces

5.1 API Endpoints

Examples:

- /api/auth/login
- /api/products
- /api/orders
- /api/invoices
- /api/payments

5.2 User Interface

- Home Page
- Product Listing Page
- Cart & Orders Dashboard
- Custom Order Form
- Admin Dashboard
- Login/Register Pages
- Invoice Viewer

5.3 Email System

- OTP Email
- Password Reset Email
- Order Status Notification
- Invoice Email

6. Non-Functional Requirements

Security

- JWT authentication
- Password hashing (bcrypt)
- OTP hashing

Performance

- Fast API response

Reliability

- Stable database operations

Usability

- Easy-to-use UI

Maintainability

- Modular code structure

Portability

- Works locally and on server

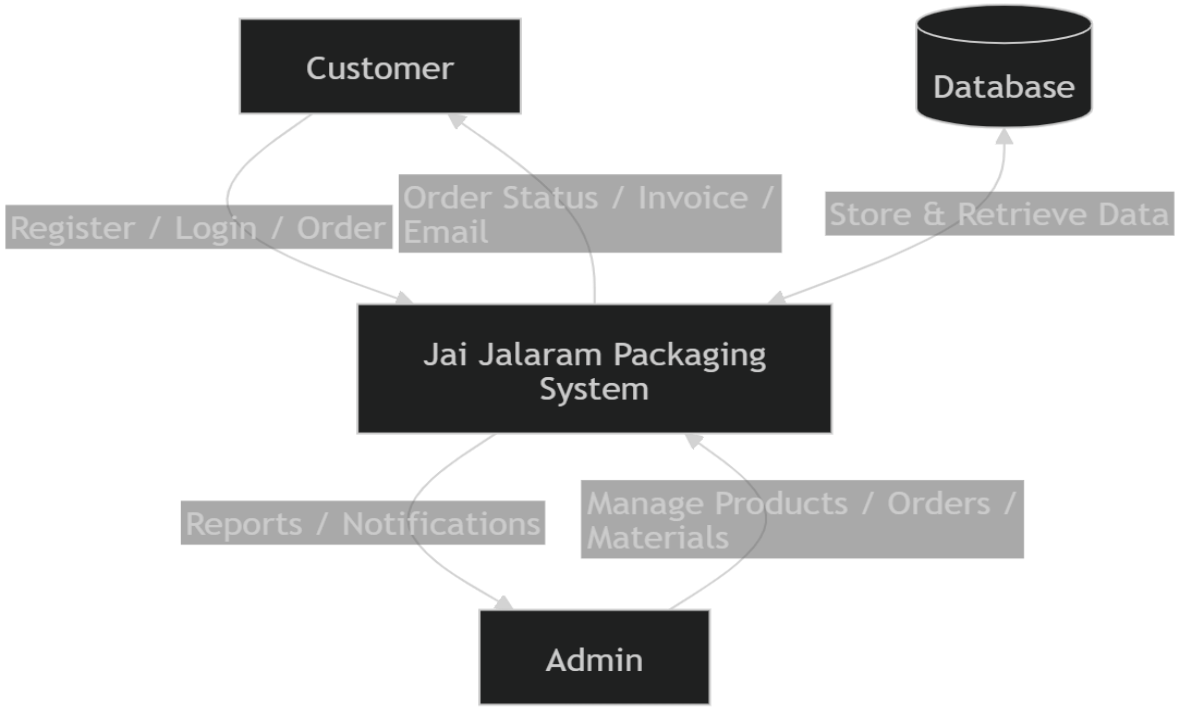
7. Environment Variables

Required variables:

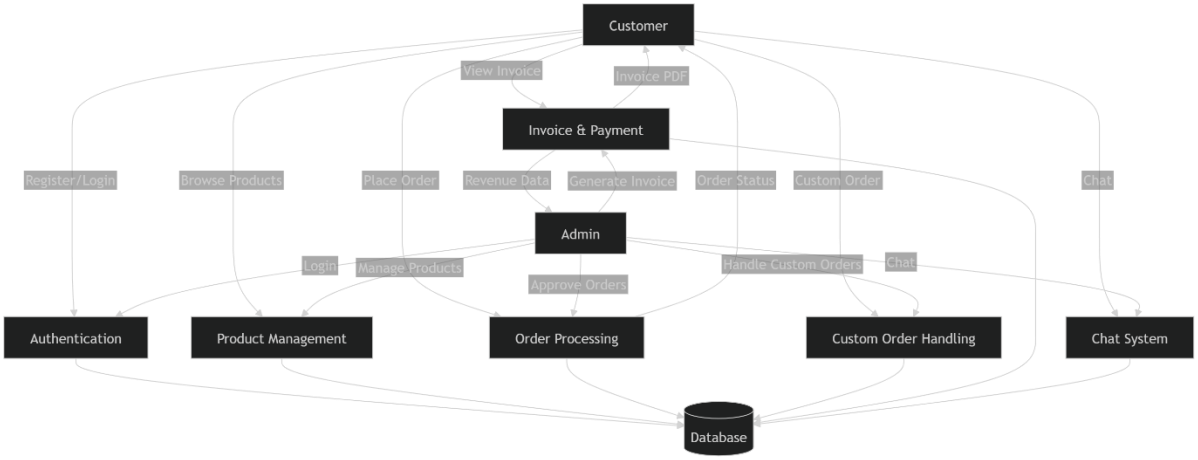
- DB_NAME, DB_USER, DB_PASS
- JWT_SECRET
- SMTP_USER, SMTP_PASS
- CLIENT_URL

DFD

Level 0

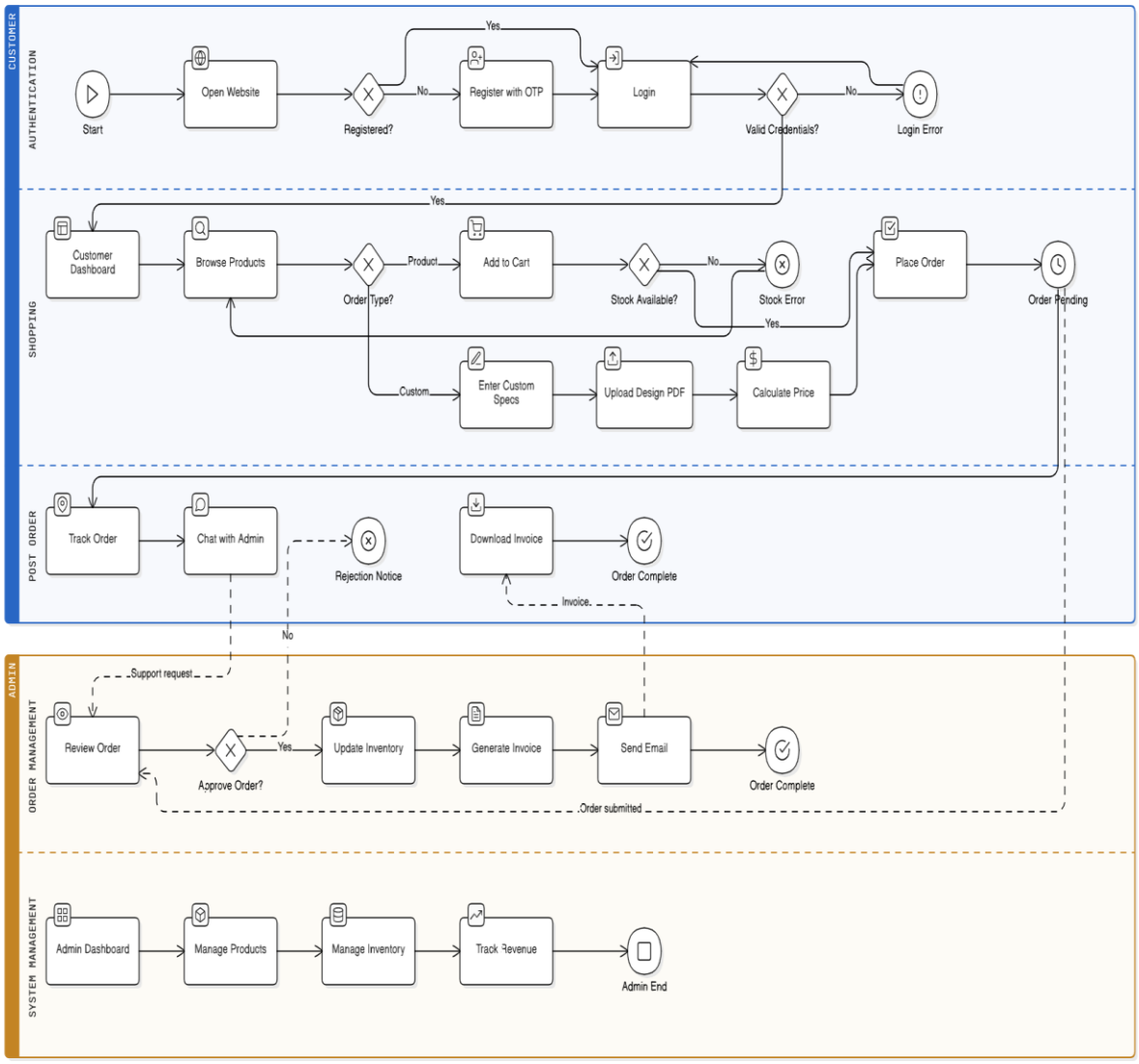
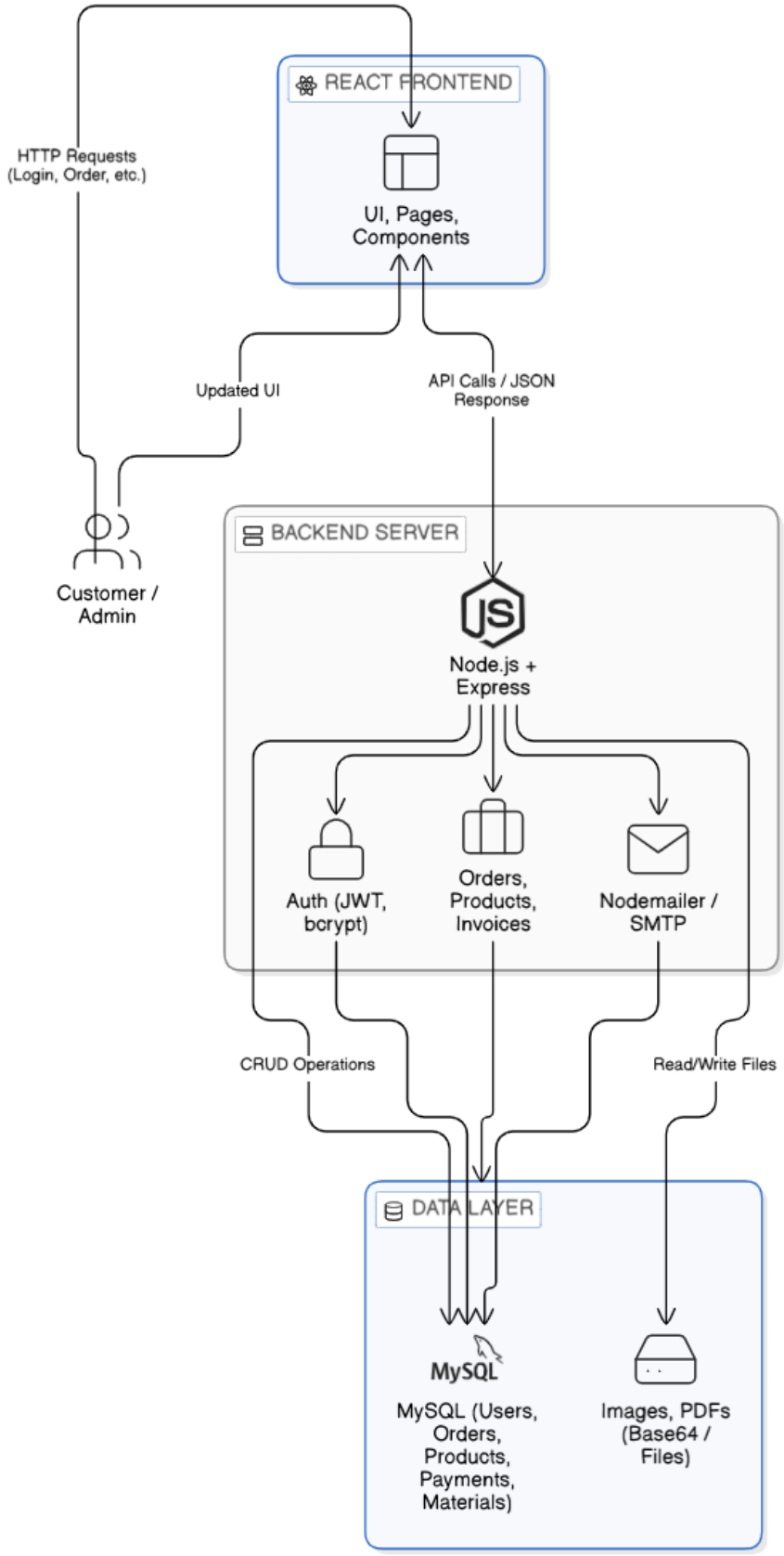


Level 1 :



Use Case Diagram

Flowchart :


CUSTOMER
AUTHENTICATION
Start
Open Website
Registered?
No
Register with OTP
Yes
Login
Valid Credentials?
No
Login Error
Yes
SHOPPING
Customer Dashboard
Browse Products
Order Type?
Product
Add to Cart
Stock Available?
No
Stock Error
Yes
Place Order
Order Pending
Custom
Enter Custom Specs
Upload Design PDF
Calculate Price
POST ORDER
Track Order
Chat with Admin
Rejection Notice
Download Invoice
Order Complete
Invoice
No
ADMIN
ORDER MANAGEMENT
Support request
Review Order
Approve Order?
Yes
Update Inventory
Generate Invoice
Send Email
Order Complete
Order submitted
SYSTEM MANAGEMENT
Admin Dashboard
Manage Products
Manage Inventory
Track Revenue
Admin End

System Architecture Diagram: